

Naimark's problem

The mathematical goal

A single unital, simple, infinite-dimensional C^* -algebra has one unitary-equivalence class of nonzero irreducible representations and a faithful tracial representation on a separable Hilbert space. The normalized trace of its CAR subalgebra extends uniquely among all states of the same generated algebra.

68 approved Cards in this selected reading scope. The separate CH consequence adds one direct Card and reuses two of these 68. The catalog contains 188 distinct Cards across the four Projects.

Scope

The principal construction concerns complex C^* -algebras and nonzero irreducible representations. Its fixed algebra and separable faithful tracial representation require no CH hypothesis. CH enters only in the statement equating the continuum with aleph one and the corresponding density-character existence result. The general reverse density obstruction also allows nonunital algebras.

68 selected Declaration Cards form this Companion, with 1,044 dependency relations, 122 displayed edges and levels 0–15. The CH consequence is a separate route reusing two density Cards.

Mathematical routes

CAR state transport and pure-state homogeneity supply one fixed shell family. Its atomic model and representation reconstruction recover the unique irreducible class, while trace extensions give a faithful separable representation of the same algebra. Density bounds yield the continuum-density counterexample; the CH equivalence is a separate consequence.

Each route shows its direct Cards and the reused prerequisites needed in the selected reading graph. Shared Cards count once within each route and once in the Catalog.

CAR foundations and canonical states

Finite matrix stages enter the CAR completion with distinct root-state and normalized-trace roles. Pointwise root-compression estimates support later residual-space arguments, while simplicity and compact-image exclusion are separate conclusions.

12 direct Cards + 0 reused prerequisites = 12 unique Cards. Shared Cards count once in this route.

- Key construction: Constructing the normalized trace on the completed CAR algebra
- Key result: Root compression becomes scalar in norm

Read this route with prerequisites

Finite-stage purification and local transport

Finite Gram matrices and matrix units purify vector states on a chosen finite stage. Corner amplification turns this transport into almost-central unitary paths, with different stage choices for protected-set and unrestricted approximation.

6 direct Cards + 8 reused prerequisites = 14 unique Cards. Shared Cards count once in this route.

- Key result: Exact vector transport along an almost central unitary path
- Key result: Cross-representation state approximation with a protected finite set

Read this route with prerequisites

Pure-state homogeneity and shell matching

Two-sided inner sequences establish pure-state homogeneity, which supplies one fixed shell family. Each non-root component uses the same approximately inner automorphism for all its shells; the generic matching theorem retains its stated hypotheses.

8 direct Cards + 15 reused prerequisites = 23 unique Cards. Shared Cards count once in this route.

- Key result: Pure-state homogeneity of the completed CAR algebra
- Key construction: Choosing the CAR shell family from proved homogeneity

Read this route with prerequisites

Direct sums of GNS representations and shell unitaries

Inequivalent pure-GNS classes form the atomic direct sum. Generic shell theorems and the CAR realization supply faithful irreducible operator models, with the required residual-line hypotheses discharged in their appropriate specialization.

7 direct Cards + 9 reused prerequisites = 16 unique Cards. Shared Cards count once in this route.

- Key result: Realizing a shell family by a faithful irreducible operator algebra
- Key result: An irreducible operator algebra constructed from projection shells

Read this route with prerequisites

Fixed subspaces and reconstruction of the generators

Decreasing transported projections determine common fixed subspaces by strong vectorwise limits. Shell sums and residual corners reconstruct the generators in arbitrary unital target representations. For a nonzero irreducible target representation, a common fixed space survives and supplies compatible selected vectors.

8 direct Cards + 12 reused prerequisites = 20 unique Cards. Shared Cards count once in this route.

- Key result: Reconstructing a represented unitary from its shells and residual corner
- Key result: An irreducible target representation has a surviving fixed space

Read this route with prerequisites

Unitary equivalence of irreducible representations

Selected GNS cyclic subspaces assemble into a surjective isometry and then a unitary equivalence of the target models. The nonunital case first proves unit preservation; shell sums are compared on their own Hilbert spaces without assuming strong continuity of an arbitrary representation.

9 direct Cards + 20 reused prerequisites = 29 unique Cards. Shared Cards count once in this route.

- Key result: Unitary equivalence without an initial unitality assumption
- Key result: The cyclic sum fills every irreducible target representation

Read this route with prerequisites

Trace extensions and faithful representations on a separable space

The CAR trace extends uniquely to a state on the same shell-generated target, and that extension is tracial. Pointed transport supplies faithful representations on separable trace spaces; representation faithfulness is distinct from faithfulness of a vector state.

9 direct Cards + 30 reused prerequisites = 39 unique Cards. Shared Cards count once in this route.

- Key result: Uniqueness among all state extensions of the CAR trace
- Key result: Faithfulness of the representation on the CAR trace Hilbert space

Read this route with prerequisites

One C^* -algebra and its representation-theoretic properties

The chosen homogeneity family defines one fixed unital simple infinite-dimensional algebra with a unique nonzero irreducible representation class. It also has a faithful representation on a separable trace space but no nonzero separable irreducible representation.

7 direct Cards + 54 reused prerequisites = 61 unique Cards. Shared Cards count once in this route.

- Key result: A separably represented C^* -algebra with no separable irreducible representation
- Key result: A simple C^* -algebra with a unique irreducible representation class

Read this route with prerequisites

Density and cardinality

A faithful separable representation bounds the carrier, while the density obstruction bounds every norm-dense subset from below. Together they give exact continuum norm density and a counterexample existence statement, keeping density, carrier cardinality and Hilbert dimension distinct.

2 direct Cards + 55 reused prerequisites = 57 unique Cards. Shared Cards count once in this route.

- Key result: Exact norm density of the fixed atomic algebra
- Key result: A Naimark counterexample of continuum norm density

Read this route with prerequisites

CH and the existence of a counterexample of norm density aleph one

The CH equivalence reuses the two continuum-density Cards and adds one direct Card. Its reverse implication applies the general density obstruction without assuming a faithful separable representation; the equivalence assumes neither CH nor its negation and makes no independence claim.

1 direct Card + 2 reused prerequisites = 3 unique Cards. Shared Cards count once in this route.

- Key result: CH and a Naimark counterexample of norm density aleph one

Read this route with prerequisites

Dependency-first reading route

Read selected Card prerequisites before their uses. Levels are recomputed from the selected reachability-preserving projection; omitted source helpers remain traceable in the source exploration.

Level 0 (6 Cards)

Level 0 The CAR algebra as a completion of finite matrix stages Names the completed CAR algebra in which finite matrix calculations can be extended by norm approximation. MathlibAnnex.CStarAlgebra.CAR.Limit Immediate Card prerequisites: None in this selected scope Used by in this scope: The product-vector state on the completed CAR algebra · Approximately inner homogeneity of pure CAR states · A finite-row average centralizes its matrix stage · The decreasing root projections in the CAR completion · Constructing the normalized trace on the completed CAR algebra · The completed CAR algebra is infinite-dimensional · A two-sided inner intertwining sequence for two states Open Declaration Card Read exact source
Level 0 The normalized trace on a finite CAR stage The normalized matrix trace provides a positive tracial functional compatible with the CAR inclusions. MathlibAnnex.CStarAlgebra.CAR.stageTrace Immediate Card prerequisites: None in this selected scope Used by in this scope: Constructing the normalized trace on the completed CAR algebra Open Declaration Card Read exact source
Level 0 Exact projection links for one approximately inner automorphism A point-norm approximation followed by close-projection conjugacy gives exact support identities for an arbitrary projection family. MathlibAnnex.CStarAlgebra.exists_shell_family_of_approximately_inner Immediate Card prerequisites: None in this selected scope Used by in this scope: Choosing the CAR shell family from proved homogeneity Open Declaration Card Read exact source
Level 0 A fixed representative family of CAR shell data One family records a transporting automorphism and exact shell links for each selected pure-state class, with an explicitly fixed root component. MathlibAnnex.CStarAlgebra.CAR.RepresentativeShellFamily Immediate Card prerequisites: None in this selected scope Used by in this scope: Choosing the CAR shell family from proved homogeneity · The matching GNS fiber retains exactly its cyclic line · Shell matching fixes the trace of transported flags · An inequivalent GNS fiber has no residual common range · Reconstructing a represented unitary from its shells and residual corner · Assembling selected GNS cyclic subspaces into an isometric source representation Open Declaration Card Read exact source

Level 0 The selected GNS representation of a pure-state class A choice of representative pure states, fixed literally at a root, gives one concrete GNS representation per equivalence class. MathlibAnnex.CStarAlgebra.PureState.selectedRepresentation Immediate Card prerequisites: None in this selected scope Used by in this scope: The matching GNS fiber retains exactly its cyclic line · Distinct GNS classes have no unitary intertwiner · The atomic direct sum of the selected CAR representations Open Declaration Card Read exact source
Level 0 The closed operator algebra generated by a representation and extra operators Places a represented algebra and a chosen family of bounded operators in one concrete closed algebra. MathlibAnnex.CStarAlgebra.AtomicConstruction.concreteTarget Immediate Card prerequisites: None in this selected scope Used by in this scope: An irreducible operator algebra constructed from projection shells · Reconstructing a represented unitary from its shells and residual corner Open Declaration Card Read exact source
Level 1 (9 Cards)
Level 1 Constructing the normalized trace on the completed CAR algebra Carries compatible bounded matrix traces through the algebraic limit, normed union and completion. MathlibAnnex.CStarAlgebra.CAR.trace Immediate Card prerequisites: The normalized trace on a finite CAR stage · The CAR algebra as a completion of finite matrix stages Used by in this scope: Shell matching fixes the trace of transported flags · Extending the CAR trace to the fixed shell target · Normalization and the trace identity determine the CAR trace Open Declaration Card Read exact source
Level 1 The product-vector state on the completed CAR algebra Compatible evaluation at the distinguished matrix coordinate extends continuously to the CAR completion. MathlibAnnex.CStarAlgebra.CAR.rootState Immediate Card prerequisites: The CAR algebra as a completion of finite matrix stages Used by in this scope: Root compression becomes scalar in norm · Purity of the completed root state Open Declaration Card Read exact source

Level 1 The completed CAR algebra is infinite-dimensional Rules out finite dimension by retaining matrix subalgebras of unbounded dimension. MathlibAnnex.CStarAlgebra.CAR.not_finiteDimensional Immediate Card prerequisites: The CAR algebra as a completion of finite matrix stages Used by in this scope: The shell-generated algebra is not an algebra of all compact operators · An irreducible CAR representation has no nonzero compact image Open Declaration Card Read exact source
Level 1 The decreasing root projections in the CAR completion The distinguished stage corners give a nested sequence of projections supporting the root state. MathlibAnnex.CStarAlgebra.CAR.rootFlag Immediate Card prerequisites: The CAR algebra as a completion of finite matrix stages Used by in this scope: Root compression becomes scalar in norm · Shell matching fixes the trace of transported flags · Reconstructing a represented unitary from its shells and residual corner Open Declaration Card Read exact source
Level 1 A finite-row average centralizes its matrix stage Turns an arbitrary element of the completed CAR algebra into one commuting with a prescribed finite matrix stage. MathlibAnnex.CStarAlgebra.CAR.ofStage_commute_rowAverage Immediate Card prerequisites: The CAR algebra as a completion of finite matrix stages Used by in this scope: Exact vector transport along an almost central unitary path · Approximating another vector state along a unitary path Open Declaration Card Read exact source
Level 1 A two-sided inner intertwining sequence for two states An inner sequence records both forward and actual inverse convergence data before any limiting automorphism is constructed. MathlibAnnex.CStarAlgebra.CAR.HasInnerIntertwiningSequence Immediate Card prerequisites: The CAR algebra as a completion of finite matrix stages Used by in this scope: Two-sided inner sequences imply pure-state homogeneity · Pure CAR states admit two-sided inner intertwining sequences Open Declaration Card Read exact source

Level 1 Approximately inner homogeneity of pure CAR states The homogeneity property asks for exact state transport by an automorphism that is locally approximable by inner automorphisms. <code>MathlibAnnex.CStarAlgebra.CAR.PureStateHomogeneity</code> Immediate Card prerequisites: The CAR algebra as a completion of finite matrix stages Used by in this scope: Two-sided inner sequences imply pure-state homogeneity Open Declaration Card Read exact source
Level 1 An irreducible operator algebra constructed from projection shells Shell partial isometries with rank-one limiting defects can be completed to unitary links that join inequivalent irreducible fibers. <code>MathlibAnnex.CStarAlgebra.AtomicConstruction.exists_irreducible_atomicShellModel</code> Immediate Card prerequisites: The closed operator algebra generated by a representation and extra operators Used by in this scope: The atomic-shell construction on selected pure-GNS fibers Open Declaration Card Read exact source
Level 1 Distinct GNS classes have no unitary intertwiner The class index makes the selected pure-GNS family pairwise unitarily inequivalent. <code>MathlibAnnex.CStarAlgebra.PureState.no_unitaryIntertwiner_selectedRepresentation</code> Immediate Card prerequisites: The selected GNS representation of a pure-state class Used by in this scope: An inequivalent GNS fiber has no residual common range · The atomic-shell construction on selected pure-GNS fibers · Assembling selected GNS cyclic subspaces into an isometric source representation Open Declaration Card Read exact source

Level 2 (6 Cards)

Level 2

Root compression becomes scalar in norm

Extends an exact rank-one compression identity on finite stages to a pointwise norm limit on the completed algebra.

`MathlibAnnex.CStarAlgebra.CAR.tendsto_norm_compressionError`

Immediate Card prerequisites: The product-vector state on the completed CAR algebra · The decreasing root projections in the CAR completion

Used by in this scope: The matching GNS fiber retains exactly its cyclic line · An inequivalent GNS fiber has no residual common range · Simplicity of the completed CAR algebra · A common root vector realizes every selected state through the generators · Assembling selected GNS cyclic subspaces into an isometric source representation

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Level 2

Purity of the completed root state

Passes extremality of the root vector states from every finite matrix stage to the completed CAR algebra.

`MathlibAnnex.CStarAlgebra.CAR.isPureState_rootState`

Immediate Card prerequisites: The product-vector state on the completed CAR algebra

Used by in this scope: Choosing the CAR shell family from proved homogeneity · The matching GNS fiber retains exactly its cyclic line · The atomic direct sum of the selected CAR representations · An inequivalent GNS fiber has no residual common range

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Level 2

Normalization and the trace identity determine the CAR trace

Proves uniqueness without assuming positivity of the competing continuous functional.

`MathlibAnnex.CStarAlgebra.CAR.eq_trace_of_apply_one_of_mul_comm`

Immediate Card prerequisites: Constructing the normalized trace on the completed CAR algebra

Used by in this scope: The fixed shell target has a unique tracial state

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Level 2

Two-sided inner sequences imply pure-state homogeneity

The forward and inverse limits close the homogeneity argument once an intertwining sequence is supplied for every pure-state pair.

`MathlibAnnex.CStarAlgebra.CAR.homogeneity_of_innerIntertwiningSequences`

Immediate Card prerequisites: Approximately inner homogeneity of pure CAR states · A two-sided inner intertwining sequence for two states

Used by in this scope: Pure-state homogeneity of the completed CAR algebra

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Level 2 The atomic-shell construction on selected pure-GNS fibers Pure-state GNS data supplies the irreducible and inequivalent fibers required by the generic shell construction. <code>MathlibAnnex.CStarAlgebra.AtomicConstruction.exists_irreducible_pureAtomicShellModel</code> Immediate Card prerequisites: Distinct GNS classes have no unitary intertwiner · An irreducible operator algebra constructed from projection shells Used by in this scope: Realizing a shell family by a faithful irreducible operator algebra Open Declaration Card Read exact source
Level 2 Shell matching fixes the trace of transported flags Recovers exact trace values from the initial and final supports of shell links. <code>MathlibAnnex.CStarAlgebra.CAR.trace_transportedFlag</code> Immediate Card prerequisites: The decreasing root projections in the CAR completion · Constructing the normalized trace on the completed CAR algebra · A fixed representative family of CAR shell data Used by in this scope: Transported flags vanish on vectors generated by a trace vector Open Declaration Card Read exact source
Level 3 (5 Cards)
Level 3 Simplicity of the completed CAR algebra Every nonzero closed two-sided ideal of the CAR algebra contains the identity. <code>MathlibAnnex.CStarAlgebra.CAR.isSimpleCStarAlgebra_limit</code> Immediate Card prerequisites: Root compression becomes scalar in norm Used by in this scope: An irreducible CAR representation has no nonzero compact image Open Declaration Card Read exact source
Level 3 The atomic direct sum of the selected CAR representations All selected pure-GNS fibers of the completed CAR algebra act together on one Hilbert direct sum. <code>MathlibAnnex.CStarAlgebra.CAR.selectedAtomicRepresentation</code> Immediate Card prerequisites: The selected GNS representation of a pure-state class · Purity of the completed root state Used by in this scope: Realizing a shell family by a faithful irreducible operator algebra · Reconstructing a represented unitary from its shells and residual corner · Assembling selected GNS cyclic subspaces into an isometric source representation Open Declaration Card Read exact source

Level 3 The matching GNS fiber retains exactly its cyclic line Identifies the residual projection of a transported CAR flag in its matching pure-state representation. MathlibAnnex.CStarAlgebra.CAR.selected_commonFixedProjection_eq_rankOne
Immediate Card prerequisites: Root compression becomes scalar in norm · The selected GNS representation of a pure-state class · Purity of the completed root state · A fixed representative family of CAR shell data Used by in this scope: The atomic common range is one embedded GNS line
Open Declaration Card Read exact source
Level 3 Transported flags vanish on vectors generated by a trace vector Turns decay of projection traces into norm convergence on each source-orbit vector. MathlibAnnex.CStarAlgebra.CAR.tendsto_transportedFlag_orbit_zero
Immediate Card prerequisites: Shell matching fixes the trace of transported flags Used by in this scope: The target represented on the CAR trace GNS space · Pointed unitary transport between trace-cyclic target representations
Open Declaration Card Read exact source
Level 3 An inequivalent GNS fiber has no residual common range Uses compression and cyclic transport to exclude fixed vectors in every other chosen pure-state class. MathlibAnnex.CStarAlgebra.CAR.selected_commonFixedProjection_eq_zero
Immediate Card prerequisites: Root compression becomes scalar in norm · Distinct GNS classes have no unitary intertwiner · Purity of the completed root state · A fixed representative family of CAR shell data Used by in this scope: The atomic common range is one embedded GNS line
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Level 4 (4 Cards)

Level 4

An irreducible CAR representation has no nonzero compact image

Simplicity and infinite dimensionality rule out compact operators coming from nonzero CAR elements.

`MathlibAnnex.CStarAlgebra.CAR.eq_zero_of_isCompactOperator_image`

Immediate Card prerequisites: Simplicity of the completed CAR algebra · The completed CAR algebra is infinite-dimensional

Used by in this scope: Lifting a corner involution to an ambient CAR unitary · Realizing a finite Gram matrix in a represented CAR corner

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Level 4

The atomic common range is one embedded GNS line

Assembles the matching and inequivalent fiber calculations in an arbitrary Hilbert direct sum.

`MathlibAnnex.CStarAlgebra.CAR.iInf_range_atomic_transportFlag_eq_span`

Immediate Card prerequisites: The matching GNS fiber retains exactly its cyclic line · An inequivalent GNS fiber has no residual common range

Used by in this scope: Every irreducible representation is unitarily equivalent to the inclusion · Realizing a shell family by a faithful irreducible operator algebra

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Level 4

Reconstructing a represented unitary from its shells and residual corner

Separates a represented generator into a strong shell sum and the exact operator between its limiting fixed spaces.

`MathlibAnnex.CStarAlgebra.CAR.exists_targetShellReconstruction`

Immediate Card prerequisites: The atomic direct sum of the selected CAR representations · The decreasing root projections in the CAR completion · A fixed representative family of CAR shell data · The closed operator algebra generated by a representation and extra operators

Used by in this scope: The target represented on the CAR trace GNS space · Pointed unitary transport between trace-cyclic target representations · An irreducible target representation has a surviving fixed space

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Level 4

Assembling selected GNS cyclic subspaces into an isometric source representation

Joins mutually orthogonal pure-state cyclic copies and controls the limiting flag projections on their sum.

`MathlibAnnex.CStarAlgebra.CAR.exists_selectedAtomicCyclicIsometry`

Immediate Card prerequisites: Root compression becomes scalar in norm · Distinct GNS classes have no unitary intertwiner · The atomic direct sum of the selected CAR representations · A fixed representative family of CAR shell data

Used by in this scope: The cyclic sum fills every irreducible target representation

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Level 5 (4 Cards)

Level 5

Lifting a corner involution to an ambient CAR unitary

A prescribed unitary involution on a represented root corner can be realized on finitely many vectors by one lifted corner exponential.

`MathlibAnnex.CStarAlgebra.CAR.exists_liftedCornerExponential_apply_eq_involution`

Immediate Card prerequisites: An irreducible CAR representation has no nonzero compact image

Used by in this scope: Exact vector transport along an almost central unitary path · Approximating another vector state along a unitary path

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Level 5

Realizing a finite Gram matrix in a represented CAR corner

The infinite-dimensional root corner contains an isometric copy of any finite family of vectors.

`MathlibAnnex.CStarAlgebra.CAR.exists_rootCornerFamily_with_gram`

Immediate Card prerequisites: An irreducible CAR representation has no nonzero compact image

Used by in this scope: Purifying a vector state on a finite CAR stage

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Level 5

Realizing a shell family by a faithful irreducible operator algebra

Constructs unitary links between pure-state summands and an irreducible algebra containing a faithful copy of CAR.

`MathlibAnnex.CStarAlgebra.CAR.exists_completedAtomicShellModel`

Immediate Card prerequisites: The atomic common range is one embedded GNS line · The atomic direct sum of the selected CAR representations · The atomic-shell construction on selected pure-GNS fibers

Used by in this scope: The ambient inclusion of the fixed shell-family target · The CAR source map into the fixed shell-family target · The C^* -algebra generated by the CAR representation and the shell unitaries

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Level 5

An irreducible target representation has a surviving fixed space

Rules out simultaneous disappearance of all limiting CAR flags by passing reduction through the reconstructed generators.

`MathlibAnnex.CStarAlgebra.CAR.exists_nonzero_targetFixedSpace`

Immediate Card prerequisites: Reconstructing a represented unitary from its shells and residual corner

Used by in this scope: A common root vector realizes every selected state through the generators

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Level 6 (5 Cards)

Level 6 Purifying a vector state on a finite CAR stage Every finite-stage restriction of a unit vector state can be realized by a unit vector in any irreducible CAR representation. MathlibAnnex.CStarAlgebra.CAR.exists_unitVector_vectorFunctional_eq_on_stage Immediate Card prerequisites: Realizing a finite Gram matrix in a represented CAR corner Used by in this scope: Approximating another vector state along a unitary path · Cross-representation state approximation with a protected finite set Open Declaration Card Read exact source
Level 6 Exact vector transport along an almost central unitary path Finite matrix-state tests ensure exact transport while protecting a prescribed finite set throughout the path. MathlibAnnex.CStarAlgebra.CAR.exists_stageTests_exact_unitary_path_apply_eq_and_conjugate_sub_norm_lt Immediate Card prerequisites: Lifting a corner involution to an ambient CAR unitary · A finite-row average centralizes its matrix stage Used by in this scope: Cross-representation state approximation with a protected finite set Open Declaration Card Read exact source
Level 6 A common root vector realizes every selected state through the generators Builds a compatible family of unit vectors from a surviving residual space and identifies their CAR vector states. MathlibAnnex.CStarAlgebra.CAR.exists_targetDefectVectorFamily Immediate Card prerequisites: Root compression becomes scalar in norm · An irreducible target representation has a surviving fixed space Used by in this scope: The cyclic sum fills every irreducible target representation Open Declaration Card Read exact source

Level 6 The CAR source map into the fixed shell-family target Restricting the codomain of the selected atomic representation gives the source map into one fixed generated target. MathlibAnnex.CStarAlgebra.CAR.shellFamilySourceHom Immediate Card prerequisites: Realizing a shell family by a faithful irreducible operator algebra Used by in this scope: The shell-generated algebra is not an algebra of all compact operators · Closed-ideal simplicity of the shell-generated algebra · Pointed unitary transport between trace-cyclic target representations · Extending the CAR trace to the fixed shell target Open Declaration Card Read exact source
Level 6 The ambient inclusion of the fixed shell-family target The same concrete target has a literal representation on the selected atomic Hilbert space. MathlibAnnex.CStarAlgebra.CAR.shellFamilyInclusion Immediate Card prerequisites: Realizing a shell family by a faithful irreducible operator algebra Used by in this scope: The ambient representation of the fixed algebra · Closed-ideal simplicity of the shell-generated algebra · Unitary equivalence without an initial unitality assumption Open Declaration Card Read exact source
Level 7 (6 Cards) Level 7 Approximating another vector state along a unitary path A fixed irreducible CAR representation can approximate any vector state on finitely many elements by moving a given unit vector. MathlibAnnex.CStarAlgebra.CAR.exists_unitary_crossRepresentation_path_approx Immediate Card prerequisites: Purifying a vector state on a finite CAR stage · Lifting a corner involution to an ambient CAR unitary · A finite-row average centralizes its matrix stage Used by in this scope: Pure CAR states admit two-sided inner intertwining sequences Open Declaration Card Read exact source
Level 7 Cross-representation state approximation with a protected finite set Finite entrywise tests allow new state approximation while an entire unitary path nearly fixes earlier data. MathlibAnnex.CStarAlgebra.CAR.exists_stageTests_crossRepresentation_path_approx Immediate Card prerequisites: Purifying a vector state on a finite CAR stage · Exact vector transport along an almost central unitary path Used by in this scope: Pure CAR states admit two-sided inner intertwining sequences Open Declaration Card Read exact source

Level 7 The shell-generated algebra is not an algebra of all compact operators Excludes an injective representation of the infinite-dimensional unital algebra onto all compact operators. <code>MathlibAnnex.CStarAlgebra.CAR.not_isCompactOperatorModel_shellFamilyTarget</code>
Immediate Card prerequisites: The CAR source map into the fixed shell-family target · The completed CAR algebra is infinite-dimensional Used by in this scope: Structural properties of one shell-generated C*-algebra · A Naimark counterexample of continuum norm density
Open Declaration Card Read exact source
Level 7 The cyclic sum fills every irreducible target representation Uses shell reconstruction and residual lines to turn an isometric CAR intertwiner into a unitary equivalence on the source. <code>MathlibAnnex.CStarAlgebra.CAR.exists_surjective_selectedAtomicCyclicIsometry</code>
Immediate Card prerequisites: A common root vector realizes every selected state through the generators · Assembling selected GNS cyclic subspaces into an isometric source representation Used by in this scope: Every irreducible representation is unitarily equivalent to the inclusion
Open Declaration Card Read exact source
Level 7 Pointed unitary transport between trace-cyclic target representations Source cyclicity and strong shell sums extend the canonical CAR orbit isometry to the whole target. <code>MathlibAnnex.CStarAlgebra.CAR.exists_pointed_unitary_of_trace_of_cyclic</code>
Immediate Card prerequisites: The CAR source map into the fixed shell-family target · Transported flags vanish on vectors generated by a trace vector · Reconstructing a represented unitary from its shells and residual corner Used by in this scope: The unique trace extension is tracial on the whole target · Uniqueness among all state extensions of the CAR trace
Open Declaration Card Read exact source
Level 7 Extending the CAR trace to the fixed shell target The source trace first becomes a state on the existing target algebra. <code>MathlibAnnex.CStarAlgebra.CAR.exists_state_extension_trace</code>
Immediate Card prerequisites: The CAR source map into the fixed shell-family target · Constructing the normalized trace on the completed CAR algebra Used by in this scope: The GNS representation of the target trace extension
Open Declaration Card Read exact source

Level 8 (3 Cards)

Level 8 Pure CAR states admit two-sided inner intertwining sequences Finite-set vector-state transport supplies explicit corrections; their summable conjugation bounds and state estimates yield the required two-sided sequence. MathlibAnnex.CStarAlgebra.CAR.hasInnerIntertwiningSequence_of_pure Immediate Card prerequisites: Approximating another vector state along a unitary path · Cross-representation state approximation with a protected finite set · A two-sided inner intertwining sequence for two states Used by in this scope: Pure-state homogeneity of the completed CAR algebra Open Declaration Card Read exact source
Level 8 Every irreducible representation is unitarily equivalent to the inclusion Extends the unitary equivalence from the CAR source to all additional generators and the norm-closed target. MathlibAnnex.CStarAlgebra.CAR.ambientInclusion_unitaryEquivalent Immediate Card prerequisites: The atomic common range is one embedded GNS line · The cyclic sum fills every irreducible target representation Used by in this scope: Closed-ideal simplicity of the shell-generated algebra · Unitary equivalence without an initial unitality assumption Open Declaration Card Read exact source
Level 8 The GNS representation of the target trace extension The chosen state on the existing target supplies its cyclic GNS representation. MathlibAnnex.CStarAlgebra.CAR.tracialRepresentation Immediate Card prerequisites: Extending the CAR trace to the fixed shell target Used by in this scope: The target represented on the CAR trace GNS space · Faithfulness of the target-state GNS representation · The unique trace extension is tracial on the whole target · Uniqueness among all state extensions of the CAR trace Open Declaration Card Read exact source

Level 9 (6 Cards)

Level 9 Pure-state homogeneity of the completed CAR algebra The constructed inner intertwining sequences discharge the supplier hypothesis and give an approximately inner transporting automorphism. MathlibAnnex.CStarAlgebra.CAR.homogeneity
Immediate Card prerequisites: Two-sided inner sequences imply pure-state homogeneity · Pure CAR states admit two-sided inner intertwining sequences Used by in this scope: Choosing the CAR shell family from proved homogeneity
Open Declaration Card Read exact source
Level 9 Unitary equivalence without an initial unitality assumption Shows why allowing nonunital input representations does not enlarge the irreducible representation class of the shell-generated algebra. MathlibAnnex.CStarAlgebra.CAR.shellFamilyInclusion_unitaryEquivalent_nonUnital
Immediate Card prerequisites: The ambient inclusion of the fixed shell-family target · Every irreducible representation is unitarily equivalent to the inclusion Used by in this scope: Structural properties of one shell-generated C^*-algebra
Open Declaration Card Read exact source
Level 9 Closed-ideal simplicity of the shell-generated algebra Uses a faithful irreducible representation in the unique unitary-equivalence class to rule out proper nonzero closed two-sided ideals. MathlibAnnex.CStarAlgebra.CAR.isSimpleCStarAlgebra_shellFamilyTarget
Immediate Card prerequisites: The ambient inclusion of the fixed shell-family target · Every irreducible representation is unitarily equivalent to the inclusion · The CAR source map into the fixed shell-family target Used by in this scope: Structural properties of one shell-generated C^*-algebra · Faithfulness of the target-state GNS representation
Open Declaration Card Read exact source
Level 9 The target represented on the CAR trace GNS space A fixed pointed unitary transports the target action onto the Hilbert space constructed from CAR alone. MathlibAnnex.CStarAlgebra.CAR.traceModelRepresentation
Immediate Card prerequisites: The GNS representation of the target trace extension · Transported flags vanish on vectors generated by a trace vector · Reconstructing a represented unitary from its shells and residual corner Used by in this scope: The fixed algebra on the CAR trace Hilbert space · Faithfulness of the representation on the CAR trace Hilbert space
Open Declaration Card Read exact source

Level 9 Uniqueness among all state extensions of the CAR trace Pointed transport identifies arbitrary state extensions before traciality is proved. MathlibAnnex.CStarAlgebra.CAR.existsUnique_state_extension_trace Immediate Card prerequisites: The GNS representation of the target trace extension · Pointed unitary transport between trace-cyclic target representations Used by in this scope: None in this selected scope Open Declaration Card Read exact source
Level 9 The unique trace extension is tracial on the whole target Unitary invariance, strong shell sums, and a closed centralizer extend the trace identity beyond the source. MathlibAnnex.CStarAlgebra.CAR.traceExtension_mul_comm Immediate Card prerequisites: The GNS representation of the target trace extension · Pointed unitary transport between trace-cyclic target representations Used by in this scope: The fixed shell target has a unique tracial state Open Declaration Card Read exact source
Level 10 (4 Cards)
Level 10 Choosing the CAR shell family from proved homogeneity CAR homogeneity and exact shell matching supply one family whose distinguished component is fixed by an explicit identity choice. MathlibAnnex.CStarAlgebra.CAR.homogeneityShellFamily Immediate Card prerequisites: Purity of the completed root state · Exact projection links for one approximately inner automorphism · A fixed representative family of CAR shell data · Pure-state homogeneity of the completed CAR algebra Used by in this scope: A simple C*-algebra with a unique irreducible representation class · The C*-algebra generated by the CAR representation and the shell unitaries Open Declaration Card Read exact source
Level 10 Structural properties of one shell-generated C*-algebra Collects the proved properties of a single generated algebra, keeping the shell-family input separate from its consequences. MathlibAnnex.CStarAlgebra.CAR.shellFamilyEndpoint Immediate Card prerequisites: The shell-generated algebra is not an algebra of all compact operators · Closed-ideal simplicity of the shell-generated algebra · Unitary equivalence without an initial unitality assumption Used by in this scope: A simple C*-algebra with a unique irreducible representation class Open Declaration Card Read exact source

Level 10 The fixed shell target has a unique tracial state Every tracial state is forced back to the CAR trace and hence to its unique extension. MathlibAnnex.CStarAlgebra.CAR.existsUnique_tracial_state Immediate Card prerequisites: Normalization and the trace identity determine the CAR trace · The unique trace extension is tracial on the whole target Used by in this scope: None in this selected scope Open Declaration Card Read exact source
Level 10 Faithfulness of the target-state GNS representation The ideal dichotomy of the same target forces the constructed nonzero representation to be injective. MathlibAnnex.CStarAlgebra.CAR.tracialRepresentation_injective Immediate Card prerequisites: The GNS representation of the target trace extension · Closed-ideal simplicity of the shell-generated algebra Used by in this scope: Faithfulness of the representation on the CAR trace Hilbert space Open Declaration Card Read exact source
Level 11 (3 Cards) Level 11 Faithfulness of the representation on the CAR trace Hilbert space Unitary conjugation preserves injectivity of the target-state representation. MathlibAnnex.CStarAlgebra.CAR.traceModelRepresentation_injective Immediate Card prerequisites: The target represented on the CAR trace GNS space · Faithfulness of the target-state GNS representation Used by in this scope: Faithfulness of the representation on the CAR trace Hilbert space Open Declaration Card Read exact source
Level 11 The C*-algebra generated by the CAR representation and the shell unitaries Defines the single algebra $\mathcal{T}$ used in the irreducible and separable faithful representations. MathlibAnnex.CStarAlgebra.CAR.AtomicCounterexampleAlgebra Immediate Card prerequisites: Choosing the CAR shell family from proved homogeneity · Realizing a shell family by a faithful irreducible operator algebra Used by in this scope: The fixed algebra on the CAR trace Hilbert space · The ambient representation of the fixed algebra Open Declaration Card Read exact source

Level 11 A simple C*-algebra with a unique irreducible representation class Establishes the properties that make the fixed algebra T a counterexample to Naimark’s problem. <code>MathlibAnnex.CStarAlgebra.CAR.atomicCounterexampleEndpoint</code> Immediate Card prerequisites: Choosing the CAR shell family from proved homogeneity · Structural properties of one shell-generated C*-algebra Used by in this scope: No nonzero irreducible representation on a separable Hilbert space Open Declaration Card Read exact source
Level 12 (2 Cards)
Level 12 The fixed algebra on the CAR trace Hilbert space Defines a representation of the same algebra T on the GNS Hilbert space of the CAR trace. <code>MathlibAnnex.CStarAlgebra.CAR.separableCounterexampleRepresentation</code> Immediate Card prerequisites: The target represented on the CAR trace GNS space · The C*-algebra generated by the CAR representation and the shell unitaries Used by in this scope: Faithfulness of the representation on the CAR trace Hilbert space Open Declaration Card Read exact source
Level 12 The ambient representation of the fixed algebra Defines the inclusion representation $\rho_0 : T \rightarrow B(\mathcal{H})$. <code>MathlibAnnex.CStarAlgebra.CAR.atomicCounterexampleRepresentation</code> Immediate Card prerequisites: The ambient inclusion of the fixed shell-family target · The C*-algebra generated by the CAR representation and the shell unitaries Used by in this scope: No nonzero irreducible representation on a separable Hilbert space Open Declaration Card Read exact source

Level 13 (2 Cards)

Level 13 Faithfulness of the representation on the CAR trace Hilbert space Proves that the representation $\tilde{\rho} : T \rightarrow B(H_\tau)$ is injective. MathlibAnnex.CStarAlgebra.CAR.separableCounterexampleRepresentation_injective
Immediate Card prerequisites: The fixed algebra on the CAR trace Hilbert space · Faithfulness of the representation on the CAR trace Hilbert space Used by in this scope: Exact norm density of the fixed atomic algebra · A separably represented C*-algebra with no separable irreducible representation
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Level 13 No nonzero irreducible representation on a separable Hilbert space Rules out irreducible representations of T on separable Hilbert spaces by the finite-dimensionality theorem for a single irreducible representation class. MathlibAnnex.CStarAlgebra.CAR.not_isIrreducible_of_separable
Immediate Card prerequisites: The ambient representation of the fixed algebra · A simple C*-algebra with a unique irreducible representation class Used by in this scope: A separably represented C*-algebra with no separable irreducible representation
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Level 14 (2 Cards)

Level 14

A separably represented C^* -algebra with no separable irreducible representation

The same algebra T has a faithful representation on a separable Hilbert space, but no nonzero irreducible representation on any separable Hilbert space.

MathlibAnnex.CStarAlgebra.CAR.atomicCounterexampleEndpoint_and_exists_separable_faithful_representation_and_no_separable_irreducible_representation

Immediate Card prerequisites: No nonzero irreducible representation on a separable Hilbert space · Faithfulness of the representation on the CAR trace Hilbert space

Used by in this scope: None in this selected scope

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Level 14

Exact norm density of the fixed atomic algebra

Combines a cardinality bound from a faithful representation on a separable Hilbert space with a lower bound for every norm-dense subset.

MathlibAnnex.CStarAlgebra.CAR.hasDensityCharacter_atomicCounterexampleAlgebra

Immediate Card prerequisites: Faithfulness of the representation on the CAR trace Hilbert space

Used by in this scope: A Naimark counterexample of continuum norm density

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Level 15 (1 Card)

Level 15

A Naimark counterexample of continuum norm density

Gives a C^* -algebra of norm density $\mathfrak{c}$ whose nonzero irreducible representations form one unitary-equivalence class, without an identification with the compact operators.

MathlibAnnex.CStarAlgebra.CAR.existsNaimarkCounterexampleOfDensity_continuum

Immediate Card prerequisites: Exact norm density of the fixed atomic algebra · The shell-generated algebra is not an algebra of all compact operators

Used by in this scope: None in this selected scope

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Source exploration and boundary inputs

The preserved 68-Card view retains the exact helper and external-provider boundary, with a separate link to the earlier 466-declaration source view. Its 68-node graph is the approved main Card scope, not a count of every source declaration.

Earlier 68-Card view and source boundary · Earlier 68-Card Project PDF · Current selected reading PDF